

Supplementary Material

Blockchain Access Control for Agricultural Edge IoT: A 61-Day Hyperledger Fabric Field Study and Implementation Postmortem (V10)

This supplement accompanies the V10 manuscript. Every table here is generated by `paper_iot/V10/analysis/s` from the same raw-data package as the main text (`run_all.sh` regenerates both).

S1. System-comparison summary

Table 1 summarises, at a coarse level, how the systems discussed in Section 2 (Related work) report their evaluation, to support the claim that laboratory or simulated short-duration testbeds dominate the literature. This table is qualitative (drawn from each cited paper’s own reported methodology) and is not generated from the raw-data package.

Table 1: Evaluation methodology of related Fabric-based / blockchain access-control systems.

System	Evaluation setting	Duration
Idrissi & Palmieri [Idrissi and Palmieri, 2024]	Constrained ARM testbed	Single-session bench
Tzenetopoulos et al. [Tzenetopoulos et al., 2022]	Edge cluster (Raspberry Pi)	Single-session bench
Pajoo et al. [Pajoo et al., 2021]	Lab testbed	Single-session bench
Ke & Park [Ke and Park, 2023]	Simulation	Single-session bench
Zaidi et al. [Zaidi et al., 2023]	Lab testbed	Single-session bench
Bandara et al. [Bandara et al., 2021]	Lab testbed	Single-session bench
This work	Working farm, 61-day field deployment plus separate controlled campaigns	61 days field + multi-day controlled campaigns

S2. Full effective-permission matrix (deployed hierarchy)

Generated directly from `chaincode/hrbac/roles.go` (the same computation used in `analysis/scripts/06_security` not transcribed).

Role	<i>ReadOwn</i>	<i>ReadZone</i>	<i>ReadAll</i>	<i>WriteSensor</i>	<i>ControlZone</i>	<i>ControlAll</i>	<i>ReadAudit</i>	<i>ManageRoles</i>	<i>IssueCrossZone</i>	<i>AdminAll</i>
Admin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gateway		Yes		Yes	Yes					
Farmer	Yes	Yes		Yes	Yes		Yes		Yes	
Agronomist	Yes	Yes			Yes		Yes		Yes	
Certifier	Yes	Yes					Yes			
SupplyChain	Yes									
Sensor				Yes						

S3. Peer-scaling topology parameters

All four configurations (4, 8, 16 and 32 logical peers) run on the same four physical Raspberry Pi hosts identified by a single `host_id` value in `raw/experiments/peer_scaling_runs.csv`. Endorsement policy, orderer topology, batch parameters and CPU/memory allocation are held constant across configurations per the deployment configuration in Table 2 of the main text; the released run manifest does not record per-run CPU, memory, disk, block height or CouchDB document-count telemetry, so pre-run system state cannot be characterised beyond what Table S7 below reports.

S4. Experimental unit by comparison

Table 2: Experimental unit used for every controlled-campaign comparison in this article.

Comparison	Unit	n
Peer-count group means / P95 (Sec. 6.4)	Live run	32 (8 per peer count)
4-vs-32-peer paired comparison (Sec. 6.1.2)	Live run, paired by counter-balancing block	8 pairs
Throughput condition/interaction model (Sec. 6.3)	Live run	100 (10 levels $\times$ 2 conditions $\times$ 5)
Energy single-channel vs. residue (Sec. 6.7)	Bench sample	150 per mode
Cryptographic timing (Sec. 6.7)	Bench sample	300 per operation
Field authorization decisions (Sec. 6.1–6.2)	Recorded decision	209,000
Authorization-boundary campaign (Sec. 6.6)	Scripted attempt	8,000

S5. Grant rate by role-operation cell

Table 3: Grant rate by role-operation cell (all cells in the trace).

Role	Operation	Grant rate (%)
Admin	ControlZone	91.39
Admin	ReadAudit	93.06
Admin	ReadEnvironmentalRecord	91.11
Admin	ReadProvenance	92.06
Admin	ReadSensorData	93.12
Admin	ReadZone	92.67
Admin	WriteSensor	92.73
Agronomist	ControlZone	92.57
Agronomist	ReadAudit	91.11
Agronomist	ReadEnvironmentalRecord	92.45
Agronomist	ReadProvenance	92.06
Agronomist	ReadSensorData	92.11
Agronomist	ReadZone	92.85
Agronomist	WriteSensor	92.57
Certifier	ControlZone	91.00
Certifier	ReadAudit	91.95
Certifier	ReadEnvironmentalRecord	92.62
Certifier	ReadProvenance	92.45
Certifier	ReadSensorData	92.29
Certifier	ReadZone	92.73
Certifier	WriteSensor	91.89
Farmer	ControlZone	90.61
Farmer	ReadAudit	92.84
Farmer	ReadEnvironmentalRecord	92.45
Farmer	ReadProvenance	91.95
Farmer	ReadSensorData	92.34
Farmer	ReadZone	92.68
Farmer	WriteSensor	92.51
Sensor	WriteSensor	100.00
SupplyChain	ControlZone	92.11
SupplyChain	ReadAudit	92.51
SupplyChain	ReadEnvironmentalRecord	92.28
SupplyChain	ReadProvenance	92.68
SupplyChain	ReadSensorData	92.62
SupplyChain	ReadZone	91.73
SupplyChain	WriteSensor	92.17

S6. Throughput by concurrency level

Concurrency	Baseline TPS	HRBAC TPS	% lower	Holm-adj. p
10	42.4	37.9	10.6	<0.001
20	51.2	46.0	10.3	<0.001
30	61.2	55.2	9.7	<0.001
40	70.5	63.0	10.7	<0.001
50	68.2	62.2	8.9	<0.001
60	66.1	60.7	8.2	<0.001
70	63.6	59.3	6.9	<0.001
80	63.4	57.4	9.4	<0.001
90	61.1	55.1	9.8	<0.001
100	59.5	54.1	9.1	<0.001

S7. All 32 peer-scaling run-level observations

Table 4: All 32 peer-scaling run-level observations.

Serial pos.	Run ID	Peers	Block (rep.)	Total lat. (ms)	Span (ms)	P95 (ms)
1	PEER-04-R1	4	1	1200.4	321.0	1276.5
2	PEER-08-R1	8	1	1141.5	319.8	1239.4
3	PEER-16-R1	16	1	1057.8	323.1	1148.7
4	PEER-32-R1	32	1	810.9	316.7	889.2
5	PEER-08-R2	8	2	1146.4	319.6	1236.0
6	PEER-16-R2	16	2	1062.2	322.5	1152.1
7	PEER-32-R2	32	2	821.7	316.5	896.0
8	PEER-04-R2	4	2	1216.1	320.9	1310.3
9	PEER-16-R3	16	3	1061.9	323.4	1148.8
10	PEER-32-R3	32	3	813.5	317.3	898.2
11	PEER-04-R3	4	3	1201.5	321.3	1289.1
12	PEER-08-R3	8	3	1154.8	320.7	1251.2
13	PEER-32-R4	32	4	812.2	317.7	910.1
14	PEER-04-R4	4	4	1200.8	321.4	1289.2
15	PEER-08-R4	8	4	1166.7	320.4	1262.5
16	PEER-16-R4	16	4	1054.3	322.6	1149.6
17	PEER-04-R5	4	5	1201.3	321.5	1291.1
18	PEER-32-R5	32	5	808.4	317.0	888.9
19	PEER-16-R5	16	5	1062.9	323.5	1153.8
20	PEER-08-R5	8	5	1145.5	320.7	1228.3
21	PEER-32-R6	32	6	813.5	317.4	912.3
22	PEER-16-R6	16	6	1060.2	322.1	1147.2
23	PEER-08-R6	8	6	1142.2	320.6	1238.4
24	PEER-04-R6	4	6	1199.8	321.3	1283.9
25	PEER-16-R7	16	7	1054.9	323.2	1142.4
26	PEER-08-R7	8	7	1142.1	320.5	1225.1

Serial pos.	Run ID	Peers	Block (rep.)	Total lat. (ms)	Span (ms)	P95 (ms)
27	PEER-04-R7	4	7	1203.9	321.5	1306.6
28	PEER-32-R7	32	7	809.2	316.7	898.9
29	PEER-08-R8	8	8	1158.9	321.1	1235.1
30	PEER-04-R8	4	8	1207.1	321.6	1307.1
31	PEER-32-R8	32	8	813.3	316.8	896.8
32	PEER-16-R8	16	8	1047.5	322.4	1143.6

S8. Counterbalancing schedule

Cross-tabulation confirming each configured peer count occupies each within-cycle position exactly twice across the two complementary Latin-square cycles (cycle = repetitions 1–4 vs. 5–8).

Peer count	Position in cycle	Cycle	Runs
4	1	1	1
8	1	1	1
16	1	1	1
32	1	1	1
4	2	1	1
8	2	1	1
16	2	1	1
32	2	1	1
4	3	1	1
8	3	1	1
16	3	1	1
32	3	1	1
4	4	1	1
8	4	1	1
16	4	1	1
32	4	1	1
4	1	2	1
8	1	2	1
16	1	2	1
32	1	2	1
4	2	2	1
8	2	2	1
16	2	2	1
32	2	2	1
4	3	2	1
8	3	2	1
16	3	2	1
32	3	2	1
4	4	2	1
8	4	2	1
16	4	2	1
32	4	2	1

S9. Revocation stage timing – withdrawn from V10

An earlier internal draft reported specific stage-duration statistics (CRL publication mean 60.4s, gossip propagation mean 4.21 min). `raw/field/revocation_stages.csv` records a single `stage_timestamp` per stage row with no paired start/end timestamp and no reliable per-episode chaining (Section 6.9.4 of the main text), so a stage *duration* cannot be recomputed from the released columns. These specific duration figures are withdrawn in V10 rather than carried forward unverified; Table 4 of the main text and `analysis/DATA_DISCREPANCIES.md` record this explicitly. Stage *counts* and the delayed-status proportion (Section 6.5) are unaffected, since they do not require a duration field.

S10. CRT recovery bounds by missing residue

Missing residue pair	Deployed bound	Corrected bound
{97, 101} (missing mod 103)	9,797	64,262
{97, 103} (missing mod 101)	9,991	64,515
{101, 103} (missing mod 97)	10,403	64,770

S11. Authorization-boundary denial-mechanism breakdown by scripted scenario

Scenario	CERTIFICATE_INVALID	OP_NOT_PERMITTED	REPLAY_DETECTED	REVOKED_CREDENTIAL	ROLE_INSUFFICIENT	ZONE_MISMATCH
brute_force_role_probe	173	164	171	180	136	176
expired_cert_reuse	149	174	165	181	165	166
replay_attack	178	157	162	189	149	165
revoked_cert_reuse	171	172	177	152	165	163
role_escalation	164	180	143	160	178	175
token_tampering	160	192	184	155	156	153
unauthorized_resource_access	171	166	164	166	176	157
zone_boundary_violation	165	168	166	157	168	176

S12. Data dictionary pointer

A field-level data dictionary for every released raw trace, including which fields are withdrawn or reinterpreted following the implementation audit, is maintained in `paper_iot/V10/analysis/VARIABLE_MAPPING.md` and `paper_iot/V10/analysis/DATA_PROVENANCE_REPORT.md`, generated alongside this supplement.

S13. Evidence needed to reduce each limitation

Table 5: Evidence that would reduce each principal limitation (Table 5 of the main text).

Limitation	What would reduce it
L1	Instrument one transaction timestamped at endorsement, ordering, validation and commit boundaries, and document the <code>rbac_overhead_ms</code> timer’s exact start/stop points in source.
L2	Re-run the peer-multiplicity campaign across genuinely distinct physical hosts (not oversubscribed logical peers on four fixed machines), and add a throughput-vs-peer-count arm.

Limitation	What would reduce it
L3	Test additional, randomized concurrency levels (not a single fixed ascending sequence) and instrument CPU, memory, disk and CouchDB service time per component to localise the saturating resource.
L4	Release a permitted-request (positive) corpus alongside the denial corpus, and encode zone, temporal, replay and identity attributes explicitly so the independent oracle can score every dimension, not only role/operation.
L5	Add an episode/correlation identifier carried through every revocation lifecycle stage, and record which post-revocation attempts were made so a success rate is estimable.
L6	Implement, tag and release the corrected chaincode, firmware and gateway verifier described in Section 4.5, then run the full performance and security campaigns against it under the same protocol as the historical measurements.

References

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